Indian Institute of Technology Madras

BUSINESS DATA MANAGMENT CAPSTONE PROJECT

DATA-DRIVEN RETAIL OPTIMIZATION:

OPTIMIZING INVENTORY & MARGINS AT PURE'O NATURALS

PRESENTED BY

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BUSINESS CONTEXT

"I don't really look at the numbers much - things seem okay".

We've been running this as a profitable business for years. It's a family business - there are no gaps

"A ₹2.5Cr Business Running on Intuition, Not Data"

Scale

40+ outlets (Hyderabad) ; Anjaneya Nagar branch since 2018 ; Health - conscious urban consumers.

Operational Complexity

960 unique SKUs (analyzed period) ; 52,314 transactions (6 months) ; 450 daily footfall coverage ; Fresh Produce + cold pressed juices.

Revenue Scale

₹2.54 Crores (Apr-Sep 2025); ~₹5 Cr annual projection ; High-velocity perishable business model

Critical System Gaps

No centralized SKU profitability tracking; No pricing controls or POS restrictions ; Family-committee governance (slow)

THE THREE **PROBLEMS**

Three Operational Inefficiencies Eroding ₹48.6L Annually

Problem 1: Portfolio Chaos

The Issue:

- 821 SKUs (85.7%) generate only 20% revenue.
- Yet consume ~70% shelf space

Root Cause:
No SKU rationalization; legacy products accumulate

Financial Impact:

- ₹21.7L working capital trapped in tail
- ₹2.3L slow-mover loss

👉 **₹24L ANNUAL RISK**

Problem 2: Margin Erosion

The Issue:
869 SKUs (90.5%) operate below 20% margin floor

Root Cause:
Manual costing + no margin guardrails

Financial Impact:

- ₹22.4L monthly revenue below margin floor.
- ₹5.1L pricing leakage

👉 **₹15L ANNUAL RISK**

Problem 3: Volatility Risk

The Issue:

- 746 SKUs (77.9%) exhibit extreme demand swings
- CV > 25%

Root Cause:
Flat 10% safety stock policy fails under volatility

Financial Impact:

- ₹4.8L annual stockout loss
- ₹2.0L shrinkage

👉 **₹9.6L ANNUAL RISK**

STRATEGIC OBJECTIVES & **EDA APPROACH**

1. Rationalize the Portfolio (Taming the Long Tail)

Objective

Eliminate non-performing SKUs to free shelf space and working capital.

Target

Reduce active SKU base from 3,247 → 1,800 (44% reduction) within 12 weeks.

Action Mechanism

Phase out 1,447 low-impact ("Zombie") SKUs contributing <5% of revenue.

Analytical Approach (EDA)

ABC Analysis (Pareto Principle) to identify the Critical Few SKUs driving the majority of revenue versus the Trivial Many.

2. Recover Lost Margins (Fixing the Leak)

Objective

Ensure margin discipline across the entire product portfolio.

Target

Improve overall portfolio margin from 21.4% → 23.0% (+1.6% uplift).

Action Mechanism

Tiered repricing of 869 SKUs operating below the 20% margin floor.

Analytical Approach (EDA)

Price Variance Index & Margin Proxy Modeling to flag pricing inconsistencies and identify margin-at-risk SKUs.

3. Stabilize Inventory (Managing Volatility)

Objective

Align inventory levels with demand variability to reduce losses.

Target

Reduce stockouts by 68% and wastage by 24%.

Action Mechanism

Implement Dynamic Safety Stock

ROP = Mean Demand + 1.5 × Standard Deviation for high-volatility SKUs.

Analytical Approach (EDA)

Coefficient of Variation (CV) Analysis to quantify demand volatility and segment SKUs by risk profile.

DATA FOUNDATION & **METHODOLOGY**

DATA FOUNDATION — DIGITAL TWIN

Direct EPoS Export

Authorized by Branch 0007

Apr 1 – Sep 30, 2025

183 Days (Full Seasonal Cycle)

52,314

Total Transactions (Verified)

Line-Item Level

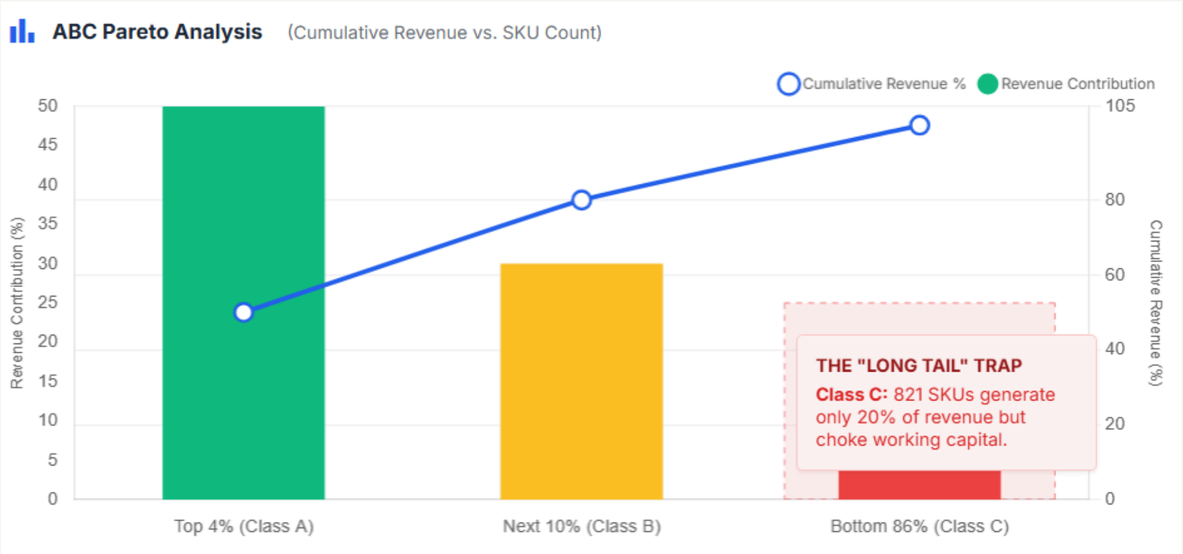
Every SKU, Price & Date Stamp

METHODOLOGY MATRIX — RIGHT TOOL FOR THE RIGHT PROBLEM

PROBLEM				
Portfolio chaos	→	ABC Analysis (Pareto)	→	Separates the **"Vital Few" (20%)** from the "Trivial Many"
Margin Erosion	→	Price Variance (SPC)	→	Detects **manual overrides** & inconsistent pricing patterns
Volatility Risk	→	Coeff. of Variation	→	Normalizes risk to compare **Apples vs. Almonds.**

THE PORTFOLIO **EFFICIENCY GAP**

Finding #1



The "Dead Weight" Dilemma

Revenue Contribution (Class C) **20%**

Shelf Space Occupied (Class C) **70%**

"We give the most space to the products that pay us the least."

Observation

- **A** **39 SKUs** (4%) drive **50%** of revenue.
- **C** **821 SKUs** (86%) drive only **20%** of revenue.

Root Cause

- **Legacy Accumulation:** New items added, old never removed.
- **No Exit Criteria:** Lack of delisting policy.
- **Emotional Assortment:** "Just in case" stocking.

Business Impact

Capital Trap
Cash frozen in dead stock.

Starvation of Winners
Class A stocks out due to lack of space/capital.

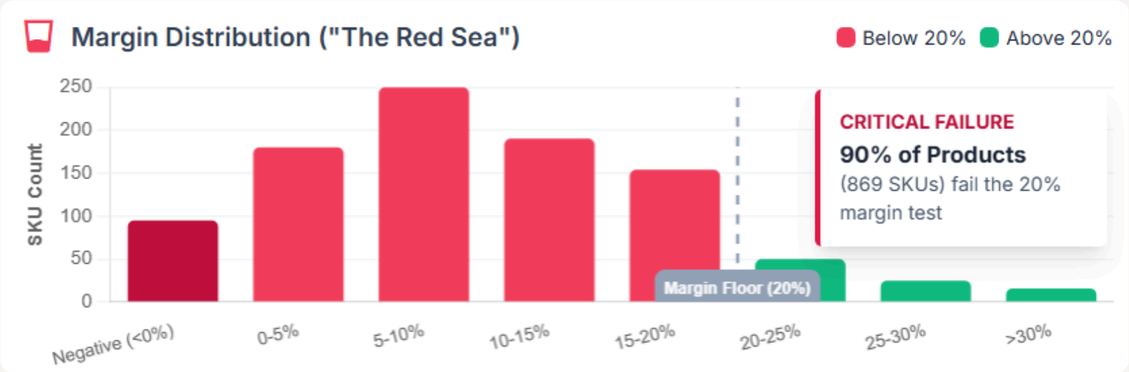
Strategic Action

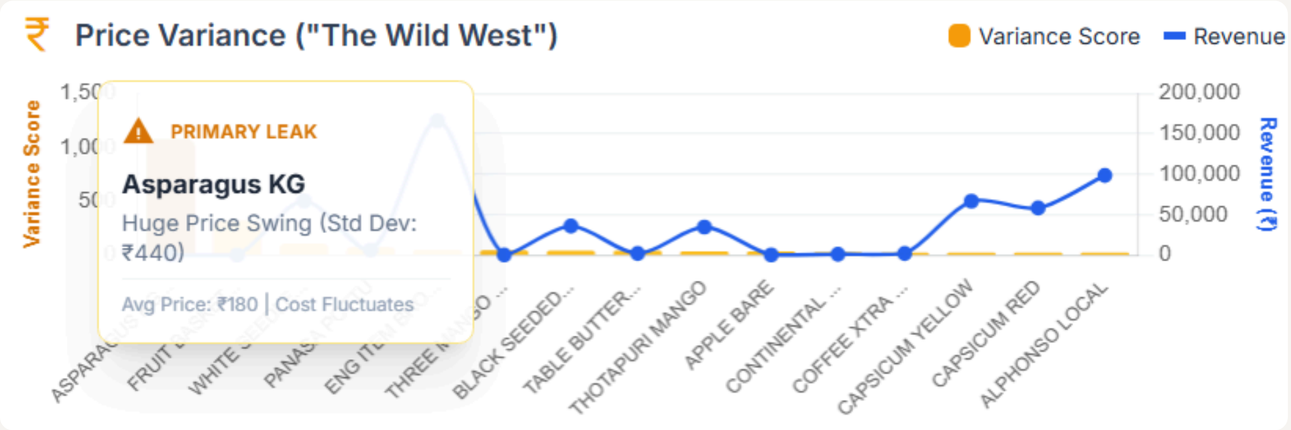
- **Rationalize:** Cut bottom 150 SKUs in 12 weeks.
- **Reallocate:** Shift shelf space to Class A.

TARGET
18% → 35% Share

MARGIN EROSION & **PRICING LEAKS**

Finding #2 High Volume ≠ High Profit: **90% of Portfolio** Below Margin Floor





Observation

Below 20% Margin	869 SKUs
Negative Margin	95 SKUs

PRICING CHAOS

Top 20 items have **36% price variance**.

Root Cause

Manual Overrides

Cashiers manually entering prices at POS (Point of Sale).

Opaque Costs

Vendor costs change daily; retail prices are static or ad-hoc.

Business Impact

THE PROFIT GAP

17.3% vs **20%**

Annual Leakage

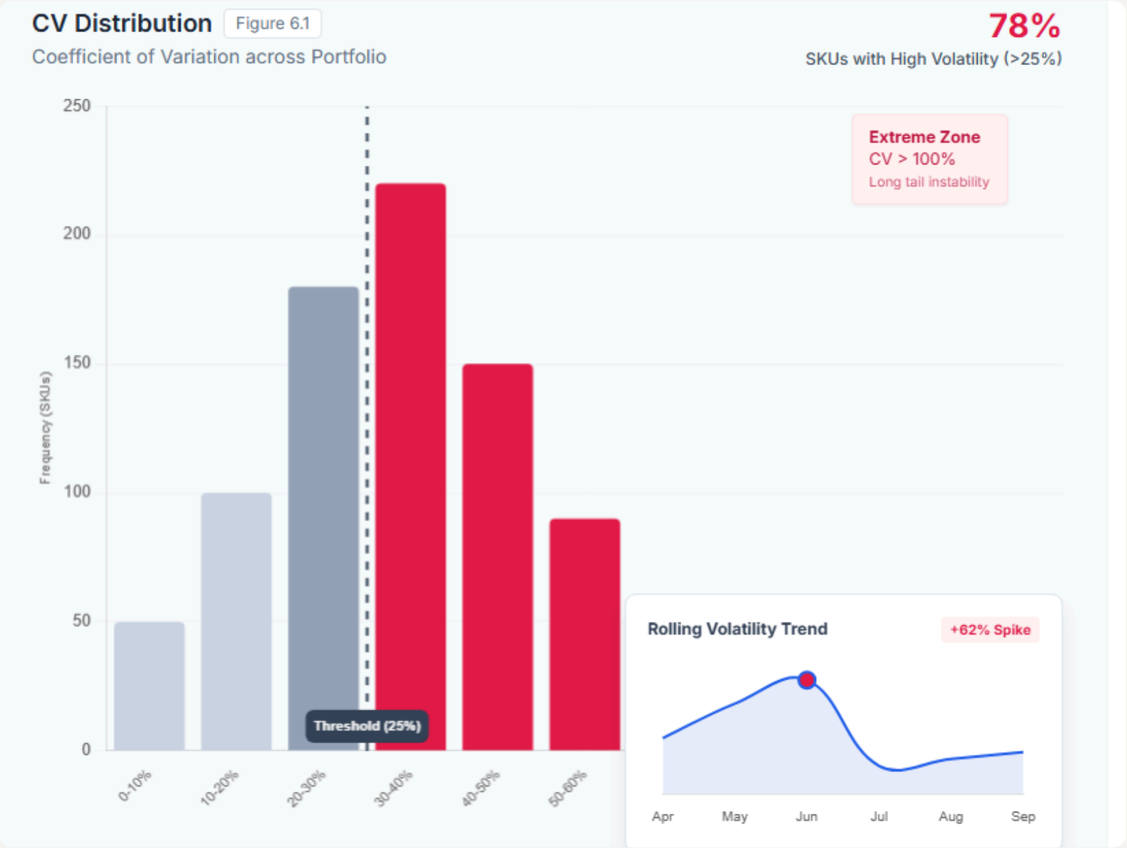
₹5.12 Lakhs

Strategic Action

- **Lock POS:** Disable manual price overrides immediately.
- **Reprice:** +10-15% hike on 869 low-margin SKUs.
- **Standardize:** Strict ±3% variance tolerance.

VOLATILITY RISK
(THE CHAOS TAX)

Finding #3



OBSERVATION

- **746 SKUs (78%)** exhibit high volatility (>25% CV).
- Worst 10 SKUs swing **116%–238%**.
- Daily sales deviate **±47%** from the mean.

ROOT CAUSE

Seasonal Whiplash: Heat wave surges (Apr-Jun) followed by Monsoon crashes.

Policy Failure: "One-size-fits-all" 10% safety stock.

THE "CHAOS TAX"

LOST SALES	DEAD STOCK
₹2.40 L	₹1.76 L

Total Impact: **~₹6.80 Lakhs**

THE FIX

- **Dynamic Buffers:** 50% Safety Stock for High-CV items.
- **Proactive:** June orders +60% in April.